

S1.3: Statistical Inference

1. Population, Sample, and Sampling Distribution

The **sampling distribution** of a statistic (e.g. the sample mean) is the probability distribution of that statistic across all possible samples of a given size.

Central Limit Theorem

If $X_1, \dots, X_n$ are IID with mean μ and variance σ^2 , then for large n ,

$$\bar{X} \approx N\left(\mu, \frac{\sigma^2}{n}\right).$$

The approximation is good for $n \geq 30$ (rule of thumb), regardless of the original distribution.

CLT is why so many statistical procedures assume approximate normality.

2. The Standard Error

The **standard error** of the sample mean is

$$\text{SE}(\bar{X}) = \frac{\sigma}{\sqrt{n}}.$$

When σ is unknown, use the sample standard deviation s in its place.

3. Hypothesis Tests for a Population Mean

Z-Test (σ Known, or Large n)

$$Z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} \sim N(0, 1) \text{ (under } H_0\text{)}$$

Critical values: ± 1.96 for 5% two-tailed; 1.645 for 5% one-tailed.

t-Test (σ Unknown, Sample Size Small)

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} \sim t_{n-1}$$

The procedure mirrors the Z-test, but critical values come from the t_{n-1} distribution.

4. Confidence Intervals

CI for μ with σ Known

$$\bar{x} \pm z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$$

CI for μ with σ Unknown (Small Sample)

$$\bar{x} \pm t_{n-1, \alpha/2} \cdot \frac{s}{\sqrt{n}}$$

Sample of 36 students: $\bar{x} = 72$, $s = 8$. Find a 95% CI for μ .

Large sample, use $z_{0.025} = 1.96$: margin = $1.96 \cdot 8/\sqrt{36} \approx 2.61$.

CI: (69.39, 74.61).

5. Type I and Type II Errors

	H_0 true	H_0 false
Reject H_0	Type I error (prob α)	Correct (power = $1 - \beta$)
Do not reject	Correct	Type II error (prob β)

- α = significance level = P(Type I) — we choose this in advance.
- β = P(Type II) — depends on the true alternative value, sample size, and α .
- **Power** of the test = $1 - \beta$ = probability of correctly rejecting a false H_0 .

Computing β

For a specific alternative $\mu = \mu_1$ and given α : find the critical region under H_0 , then compute the probability of the test statistic *falling outside* the critical region when $\mu = \mu_1$.

One-tailed Z-test of $H_0 : \mu = 10$ vs. $H_1 : \mu > 10$, $\sigma = 2$, $n = 16$, $\alpha = 0.05$. Find β at $\mu = 11$.

Standard error = $2/4 = 0.5$. Critical value: $\bar{x} > 10 + 1.645(0.5) = 10.823$.

Under $\mu = 11$: $P(\bar{X} \leq 10.823) = P(Z \leq (10.823 - 11)/0.5) = P(Z \leq -0.354) \approx 0.362$.

So $\beta \approx 0.362$; power ≈ 0.638 .

6. Sample Size Determination

To achieve a margin of error E in a CI at confidence $1 - \alpha$:

$$n \geq \left(\frac{z_{\alpha/2} \sigma}{E} \right)^2$$

7. Exam Style Strategy

- Identify whether σ is known (use Z) or unknown (use t).
- State null/alternative hypotheses symbolically.
- For CIs, state confidence level explicitly.

- For power/Type II calculations, draw the two overlapping distributions — it clarifies which tail you're computing.